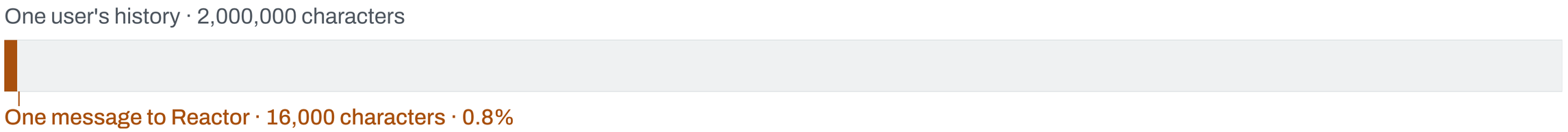
Dealer Handshake

Reactor answers. Code does the rest.

Find the two sentences that matter in two million characters of conversation.

One user, 200 conversations, one question. The answer is two sentences.



8%

of questions have no answer in the history. A guess scores as wrong.

6

cat names in one history. Only one is the user's cat today.

146

test questions · 15 users · 200 conversations each

Only Reactor writes the answer.

Every answer cites its conversations.

Not in the history? Abstain.

Retrieval found it. Reactor refused to read it.

18 answerable dev questions · every needed conversation already in the packet



- 11 correct
- 6 refused, with the answer in the packet
- 1 wrong count

33%

false refusals, with the answer already in front of it

LONG CONTEXT

16,000 characters per message. Twice the context: +25% time, 0 fewer refusals.

TIME

29–115 s per answer, one question at a time. 3 hours per pass.

NUMBERS

Counts from prose: said 1, then 2. The answer was 3.

CALCULATIONS

Refused to subtract two dates. The answer was 6 days.

CONSISTENCY

Same message, two runs: a refusal, then the right answer.

CITATIONS

Cites only the latest update. Evidence F1 0.67.

Give Reactor less.
Only the 16,000 characters that matter.



Exact. Every needed conversation in the packet 96.7% of the time.

Fast. 61 s per answer in the final run.

Local. No other LLM anywhere. Two small models on our CPU.

We played around. We stopped here.



DROPPED

Twice the context: +25% time, the same refusals.

DROPPED

20 sources instead of 12: the answer sentence got cut.

KEPT

Dates as day numbers, computed by code. Subtraction, not reasoning.

DROPPED

An event list for counting: Reactor counted the list, not the excerpts.

KEPT

Ettin instead of MiniLM: 86.8% → 89.5% of packets complete.

KEPT

Ownership cues: “your sister has 3 beehives” is not yours.

BM25. The classic search formula.

PROS

The most reliable signal for names, titles and numbers.

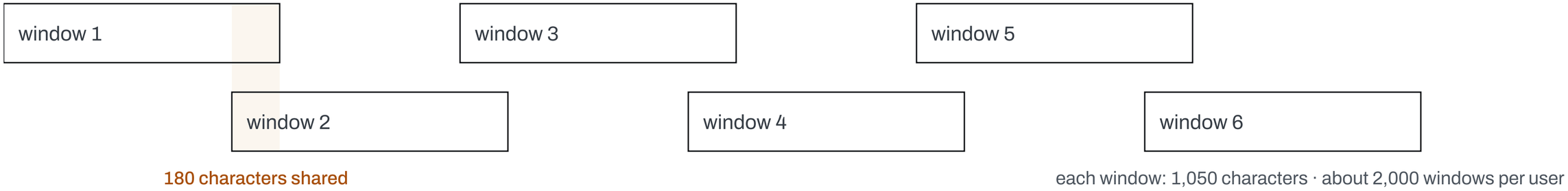
CONS

Misses rephrasing: “director” vs “my title”.

WHAT WE DO

Cut every conversation into overlapping windows, so nothing falls between two of them.

one conversation · about 8,000 characters



BGE embeddings. Search by meaning.

PROS

Finds the same thing said in other words. What BM25 misses.

CONS

“My cat” and “the neighbour's cat” look almost the same.

WHAT WE DO

The same overlapping windows, embedded by a 33M-parameter model on our own CPU.



Ettin reranker. A relevance score.

PROS

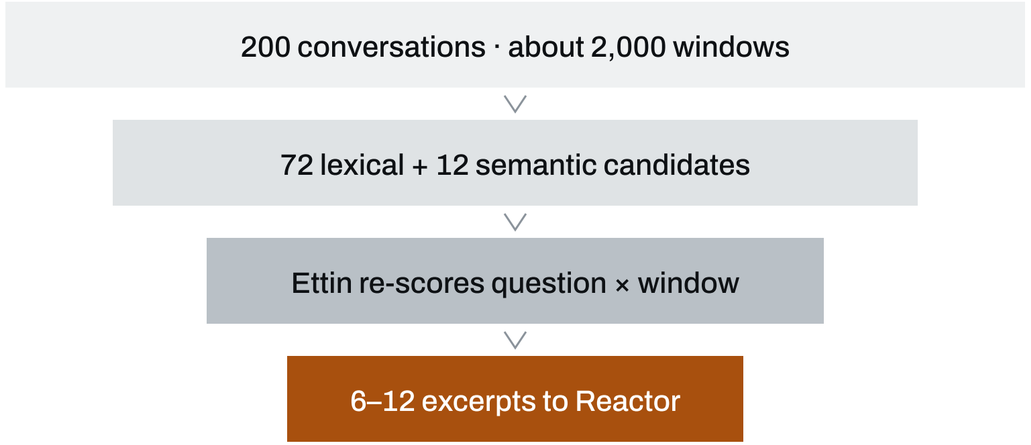
Reads the question and the window together. Tells my cat from the neighbour's.

CONS

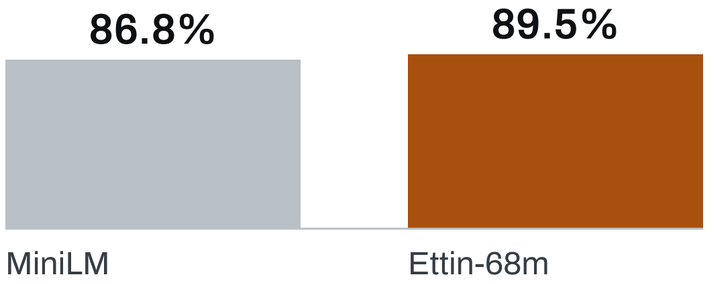
Expensive. So it re-checks only the top candidates.

WHAT WE DO

Score the shortlist from BM25 and BGE, never the whole history.



packets with every needed conversation · 500 conversations per user · n = 38



RRF. Three lists, one ranking.

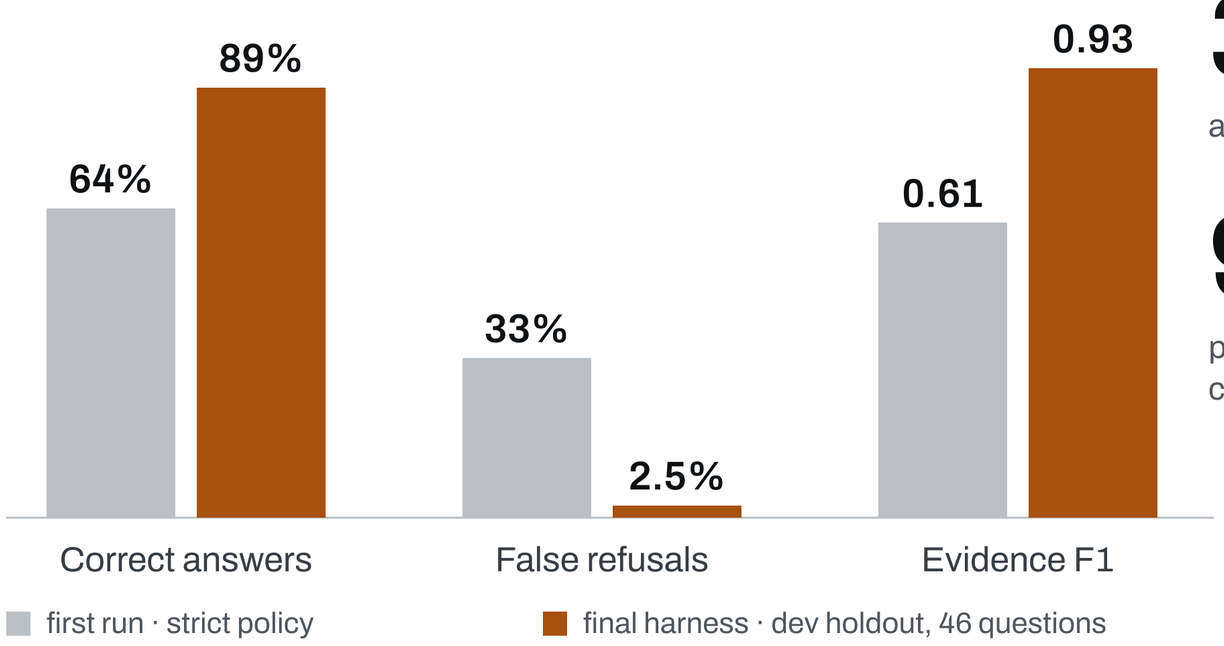
Unsupervised. Nothing to train. Each list votes by rank, not by score.

$$\text{score} = 1/(12+\text{rank}_{\text{reranker}}) + 0.5/(12+\text{rank}_{\text{BGE}}) + 0.25/(12+\text{rank}_{\text{BM25}})$$

	rank by reranker	rank by BGE	rank by BM25	score	
Excerpt A	#1	#3	#5	0.125	
Excerpt B	#2	#1	#1	**0.129**	

B first: consistent across three lists beats a single first place. The 12 keeps one list from dominating.

From 64% to 89%.



39 / 40
answerable questions answered

5 / 6
unanswerable questions refused

96.7%
packets with every needed conversation

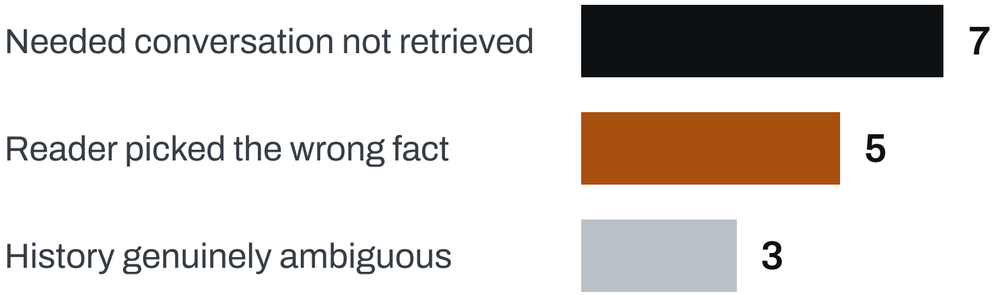
7 / 7
owner-swap decoys refused

TEST SUBMISSION 146 questions 131 answers 15 abstentions 2 h 54 min 61 s per call

hand audit of 129 answers: 112 correct · 87%

Where the remaining errors are.

15 TEST ANSWERS THE AUDIT FLAGGED · BY CAUSE



Most misses share no words with the question: “job title” vs “promotion… Director”.

RETRIEVAL

Expand the question with synonyms: job title → promotion, position, director.

COUNTING

Confirm candidates one by one; code adds them up.

NUMBERS

Ask twice. Take the majority.

CITATIONS

Cite the old value and the new one.

ABSTENTION

Two policies must refuse before we abstain.

SPEED

A second Reactor agent: half the time per pass.

Dealer Handshake

Any questions?

CODE · PREDICTIONS · EXPERIMENTS

github.com/qavitalii/reactor_research

DEMO

d7evc6l18pol2.cloudfront.net